Colligative Properties

Colligative properties are properties that depend on the concentration of a solute but not on its identity.

Definition: A colligative property may be defined as one which depends on the number of particles in solution and not in any way on the size or chemical nature of the particles.

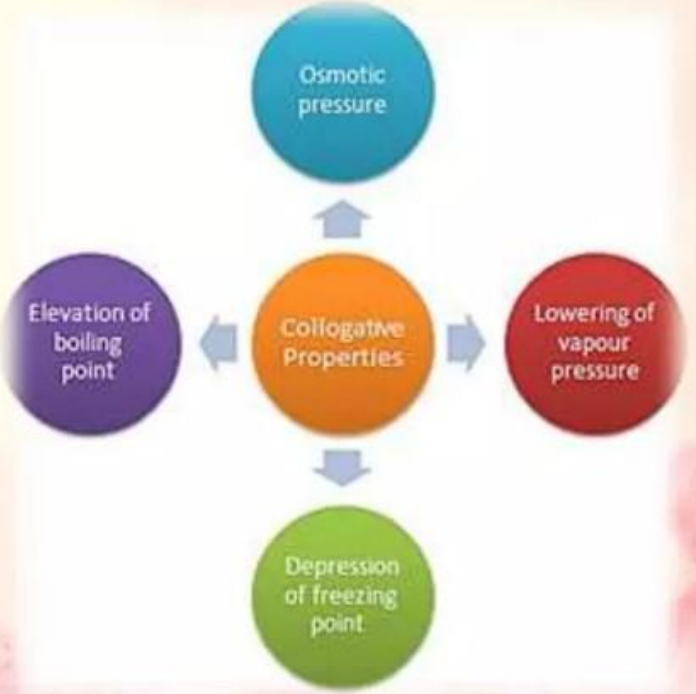
The four principal colligative properties are

(1) Lowering of the Vapour Pressure

(2) Elevation of the Boiling Point

(3) Depression of the Freezing Point

(4) Osmotic Pressure



The essential feature of these properties is that they depend *only* on the number of solute particles present in solution. Being closely related to each other through a common explanation, these have been grouped together under the class name ***Colligative Properties*** (Greek *colligatus* = Collected together).

Importance

a) Molecular mass of substances can be determined.
b) Whether a solution is iso-osmotic or not can be found.
c) The behavior of solution of electrolytes can be understood.
d) The osmotic properties of body fluids such as lacrimal fluids and blood can be evaluated.
e) Isotonic solutions can be prepared.

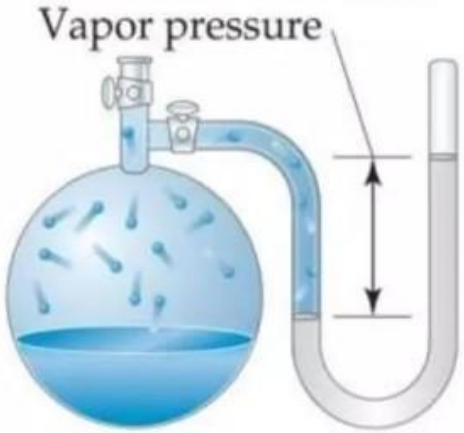
Colligative Properties – The lowering of Vapor Pressure

The lowering of vapor pressure is a decrease in the vapor pressure of a solution when a nonvolatile solute is dissolved in a solvent:

The solute particles occupy some of the surface area that would otherwise be available for solvent molecules to evaporate. This results in less solvent being able to evaporate, which lowers the vapor pressure.

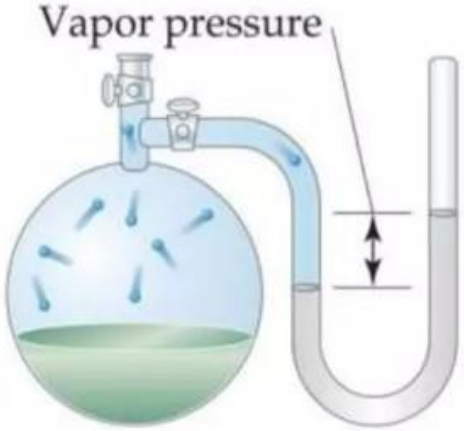
- A solvent in a closed container reaches a state of dynamic equilibrium.
- The pressure exerted by the vapor in the headspace is referred to as the vapor pressure of the solvent.
- The addition of any nonvolatile solute (one with no measurable vapor pressure) to any solvent reduces the vapor pressure of the solvent.
- Nonvolatile solutes reduce the ability of the surface solvent molecules to escape the liquid.
- Vapor pressure is reduced.
- The extent of vapor pressure lowering depends on the amount of solute.
- Raoult’s Law quantifies the amount of vapor pressure lowering that is observed.

Lowering of vapour pressure



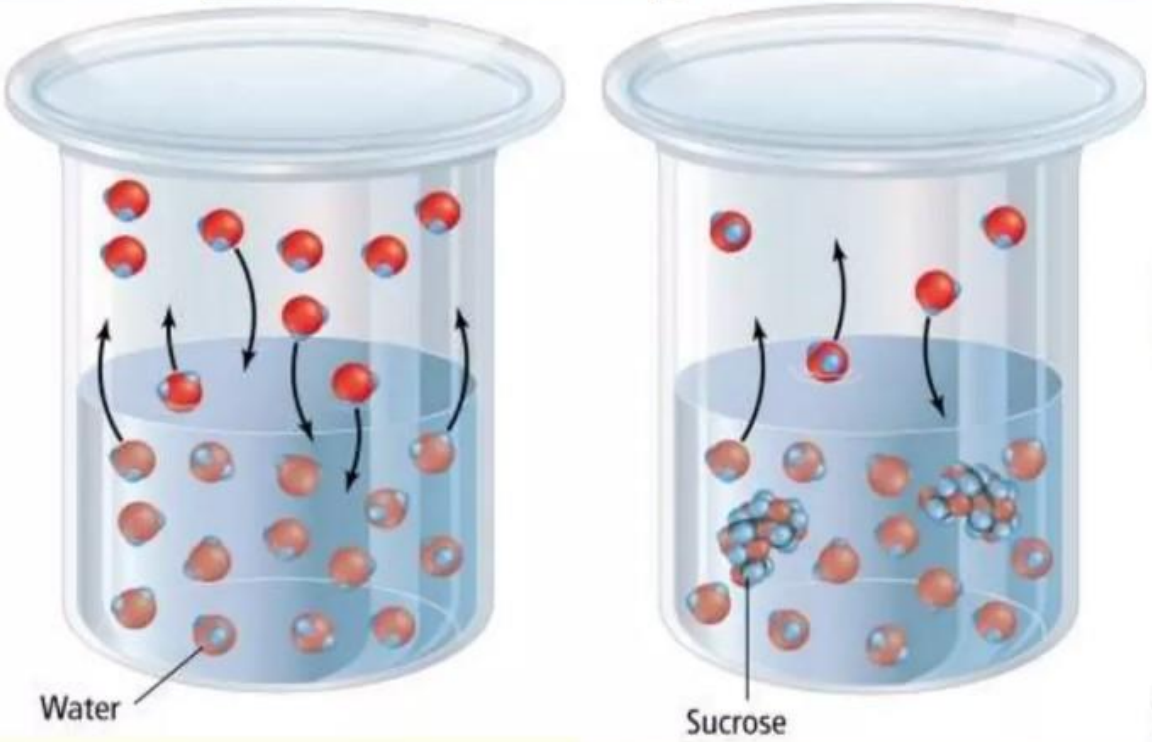
Solvent alone

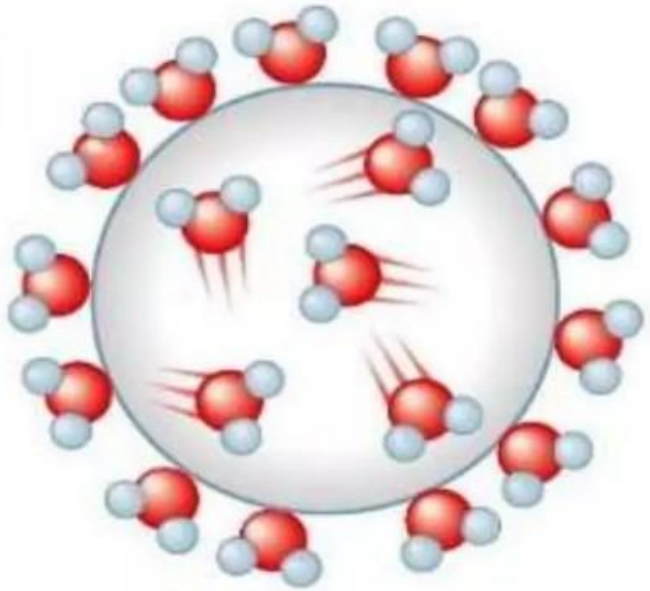
(a)



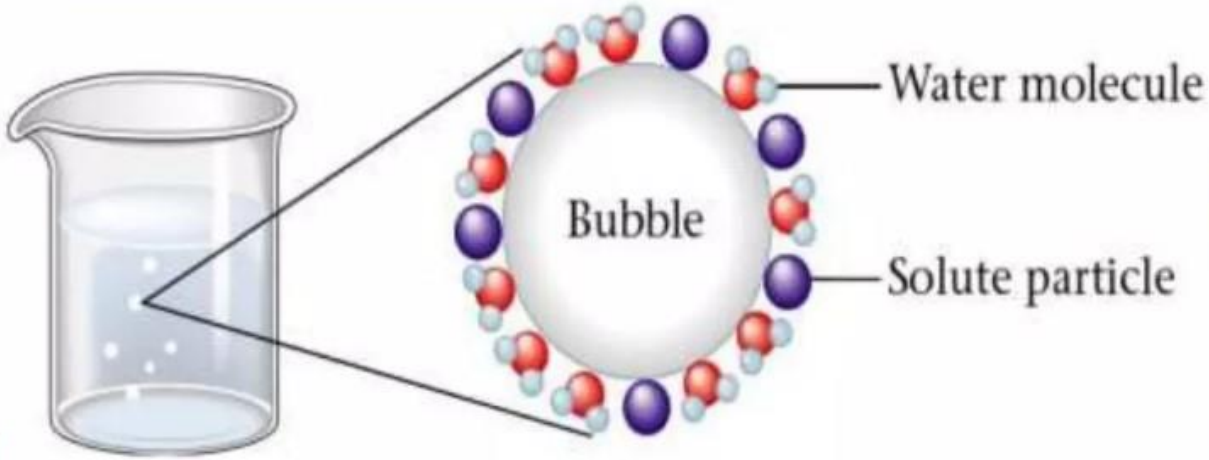
Solvent + solute

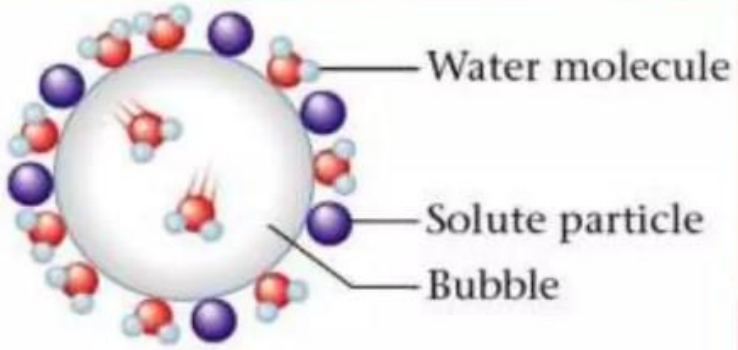
(b)





(a) Pure water





(b) Solution (contains solute)

Lowering of Vapour Pressure: Raoult's Law

The vapour pressure of a pure solvent is decreased when a non-volatile solute is dissolved in it. Raoult (1886) gave an empirical relation, connecting the relative lowering of vapour pressure and the concentration of the solute in solution. This is now referred to as the **Raoult's Law.**

It states that: **the relative lowering of the vapour pressure of a dilute solution is equal to the mole fraction of the solute present in dilute solution.**

If p is the vapour pressure of the solvent and p_s that of the solution, the lowering of vapour pressure is $(p - p_s)$. This lowering of vapour pressure relative to the vapour pressure of the pure solvent is termed the **Relative lowering of Vapour pressure.**

Thus,

Relative Lowering of Vapour Pressure $= \dfrac{p - p_s}{p}$

Therefore, Raoult's Law can be expressed mathematically in the form: $\dfrac{p - p_s}{p} = \dfrac{n}{n + N}$

where n = number of moles or molecules of solute, N = number of moles or molecules of solvent.

Derivation of Raoult's Law

Let, p is the vapour pressure of the solvent and p_s that of the solution, the vapor pressu re of the solution is directly proportional to the mole fraction of the solvent. *The vapor pressure of the* ***solution is,*** *therefore, determined by the number of molecules of the solvent present at any time in the surface which is proportional to the mole fraction.*

That is, $p_s \propto \frac{N}{n+N}$

Where N = moles of solvent and n = moles of solute.

$$p_s = k\frac{N}{n+N} \text{--------}(1)$$

Where, k =proportionality factor.

In case of pure solvent, n=0

And hence mole fraction of solvent $= \frac{N}{n+N} = \frac{N}{0+N} = 1$

Now from equation (1), the vapor pressure of the solvent $p = k$

Therefore the equation (1) assumes the form

This is Raoult's Law.

$$p_s = p\frac{N}{n+N}$$

$$\frac{p_s}{p} = \frac{N}{n+N}$$

$$1 - \frac{p_s}{p} = 1 - \frac{N}{n+N}$$

$$\frac{p - p_s}{p} = \frac{n}{n+N}$$

Determination of Molecular Mass from Vapour Pressure Lowering

The molecular mass of a non-volatile solute can be determined by measuring the lowering of vapour pressure ($p - p_s$) produced by dissolving a known weight of it in a known weight of the solvent. If in a determination *w* grams of solute is dissolved in *W* grams of the solvent, *m* and *M* are molecular masses of the solute and solvent respectively, we have:

No. of Moles of solute $(n) = \frac{w}{m}$ and No. of Moles of solvent $(N) = \frac{W}{M}$

We know that, Raoult's Law $\frac{p - p_s}{P} = \frac{n}{n + N}$

Substituting these values in the Raoult's law Equation, $\frac{p - p_s}{P} = \frac{w/m}{w/m + W/M}$ ----------(1)

Since for very dilute solution, the number of moles (molecules) of solute (w/m), is very small, it can be neglected in the denominator.

The equation (1) can now be written as $\frac{p - p_s}{P} = \frac{wM}{mW}$ ----------(2)

Knowing the experimental value of $p - p_s/p$, and the molecular mass of the solvent (M), the molecular weight of solute (m) can be calculated from (1) or (2).

- ✓ Ideal solutions are those that obey Raoult's Law.
- ✓ Real solutions show approximately ideal behavior when:
  - The solution concentration is low
  - The solute and solvent have similarly sized molecules
  - The solute and solvent have similar types of intermolecular forces.

- Raoult's Law breaks down when solvent-solvent and solute-solute intermolecular forces of attraction are much stronger or weaker than solute-solvent intermolecular forces.

Colligative Properties – Vapor Pressure

- Raoult's Law:

$$P_A = X_A P^o_A$$

where P_A = partial pressure of the solvent (A) vapor above the solution (ie with the solute)

X_A = mole fraction of the solvent (A)

P^o_A = vapor pressure of the pure solvent (A)

Example: The vapor pressure of pure water at 110°C is 1070 torr. A solution of ethylene glycol and water has a vapor pressure of 1.10 atm at the same temperature.

What is the mole fraction of ethylene glycol in the solution?

Explanation

To determine the mole fraction of ethylene glycol in a solution, we can use Raoult's Law. Raoult's Law states that the vapor pressure of a solution is directly proportional to the mole fraction of the solvent (water, in this case) and its corresponding vapor pressure when pure.

1.Given Information:

1 atm =760 torr

1. Vapor pressure of pure water at 110 °C, P° = 1070 torr.
2. Vapor pressure of the solution at 110 °C, P = 1.00 atm.

2.First, convert the vapor pressure of pure water from torr to atm: $P°$=760 torr/atm1070 torr=1.4079 atm

3.Use Raoult's Law:

1. According to Raoult's Law, we can calculate the mole fraction of the solvent: $P=xsolvent\times P°$ Here, P is the vapor pressure of the solution,$xsolvent$ is the mole fraction of the solvent (water), and $P°$ is the vapor pressure of the pure solvent.

4.Substituting the known values: 1.00 atm=$xsolvent$×1.4079 atm

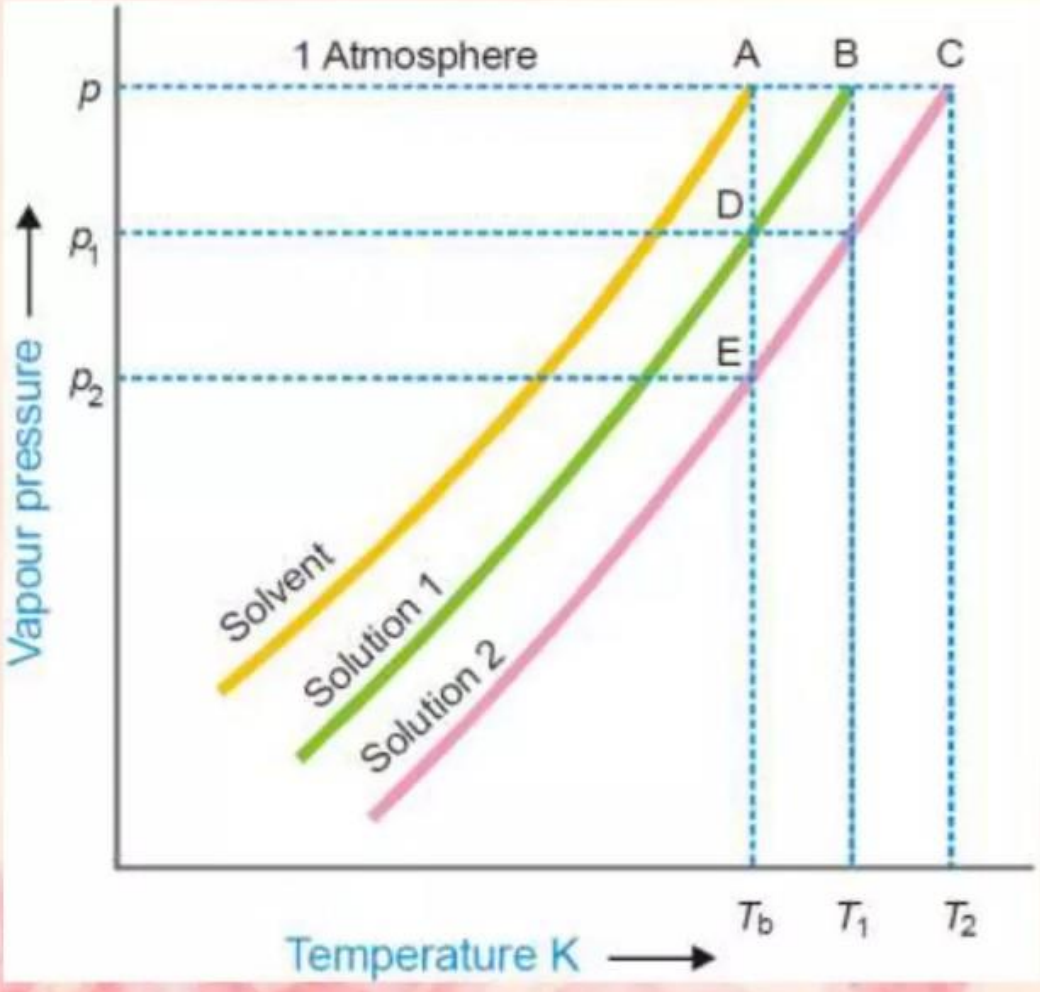
5.Now, solve for $xsolvent$: $xsolvent$=1.4079 atm1.00 atm≈0.709

6.Finding the mole fraction of ethylene glycol:

1. The total mole fractions of components in a solution equals 1: $xsolvent+xsolute=1$ Thus, $xsolute$ (mole fraction of ethylene glycol) can be calculated as: $xsolute=1-xsolvent$ $xsolute=1-0.709\approx0.291$

7.Conclusion: The mole fraction of ethylene glycol in the solution is approximately 0.291.

Elevation of Boiling Point



When a liquid is heated, its vapour pressure rises and when it equals the atmospheric pressure, the liquid boils. The addition of a non-volatile solute lowers the vapour pressure and consequently elevates the boiling point *as more heat is needed to supply additional kinetic energy to raise the vapour pressure to atmospheric pressure*. It is Called boiling-point elevation.

If T_b is the boiling point of the pure solvent and T is the boiling point of the solution of a nonelectrolyte in that solvent, the difference in the boiling points (ΔT_b) is called the elevation of boiling point.$T - T_b = \Delta T_b$

For dilute solutions, the curves *BD* and *CE* are parallel and straight lines approximately. Therefore for similar triangles *ACE* and *ABD*, we have

$$\frac{AB}{AC} = \frac{AD}{AE}$$

$$\text{or,} \frac{T_1 - T_b}{T_2 - T_b} = \frac{p - p_1}{p - p_2}$$

Where $p - p_1$ and $p - p_2$ are lowering of vapour pressure for solution 1 and solution 2 respectively.

Hence the elevation of boiling point is directly proportional to the lowering of vapour pressure.or $\Delta T_b \propto (p - p_s)$.

Raoult's Law of boiling point elevation

(i) The elevation of boiling point of a solution is proportional to its molal concentration i.e. to its molality, m.

$\Delta T_b \propto m$

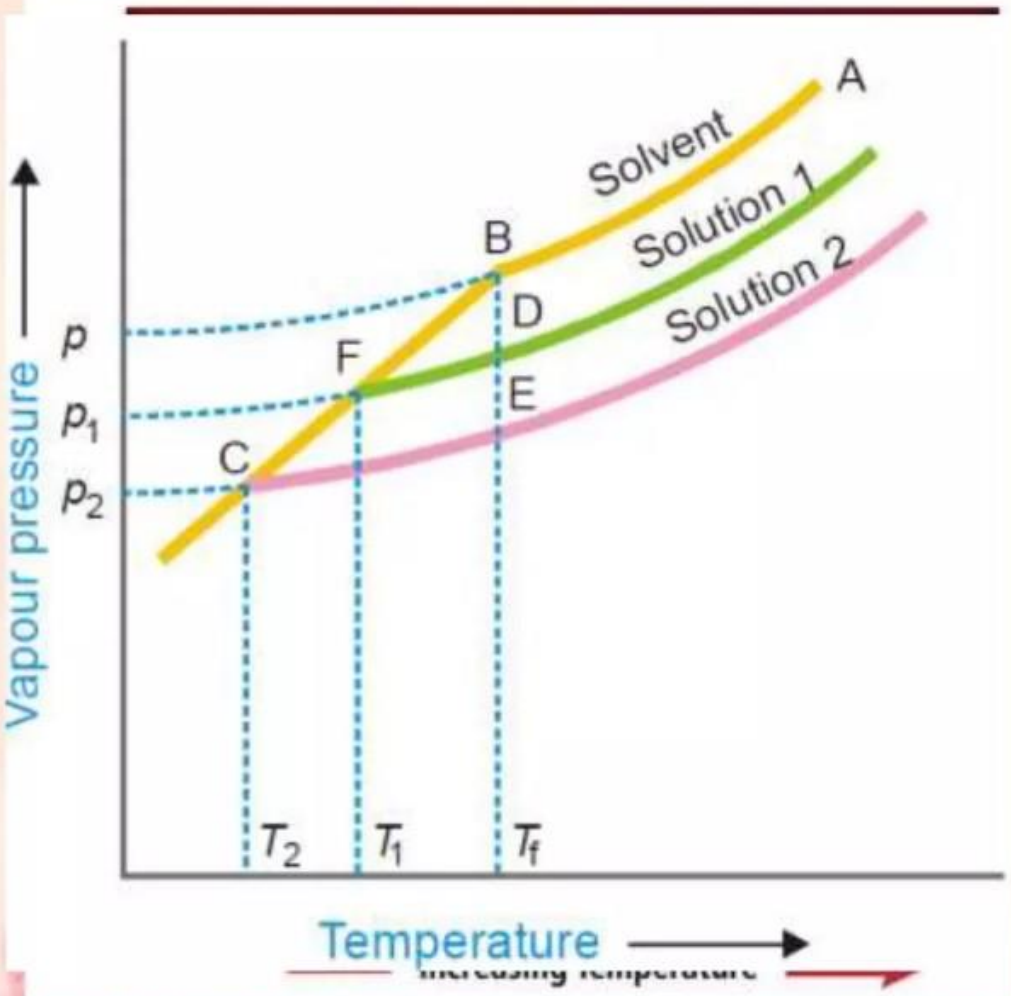
Or, $\Delta T_b = K_b . m$ where K is known as Boiling point constant, or **Ebbulioscopic constant** or **Molal elevation constant.**

When m=1, then $T_b = K_b$

So, molal elevation constant may be defined as boiling point elevation produced when 1 mole of solute is dissolved in one kg (1000 g) of the solvent.

(ii) Equimolecular quantities of different substances dissolved in the same quantity of a particular solvent raise its boiling point to the same extent.

DEPRESSION OF FREEZING POINT



The freezing point of a solution is always lower than that of the pure solvent.

The difference of the freezing point of the pure solvent and the solution is referred to as the **Depression of freezing point.** It is represented by the symbol ΔT or ΔT_f . And Depression of freezing point. Is $T_f - T_1 = \Delta T$

Derivation

The vapour pressure curve of a solution (*solution* 1) of a non-volatile solute meets the freezing point curve at *F*, indicating the freezing point of the solution, T_1. Addition of more solute causes a further lowering of freezing point to T_2. Here the freezing point of pure solvent, T_f.

For dilute solutions FD and CE are approximately parallel straight lines and BC is also a straight line. Since the triangles BDF and BEC are similar, thus

$$\frac{DF}{EC} = \frac{BD}{BE}$$

$$\frac{T_f - T_1}{T_f - T_2} = \frac{p - p_1}{p - p_2}$$

where P_1 and P_2 are vapour pressure of solution 1 and solution 2 respectively. Hence depression of freezing point is directly proporti onal to the lowering of vapour pressure.

or $\Delta T_f \propto (p - p_s)$.

RAOULT'S LAW OF DEPRESSION OF FREEZING POINT

(i) The depression of freezing point of a solution is proportional to its molal concentration i.e. to its molality, m.

$\Delta T_f \propto m$

$\Delta T_f = K_f . m$ **where K_f is known as molal depression of freezing point constant or cryoscopic constant.**

When m=1, then $T_b = K_b$.

So, cryoscopic constant may be defined as freezing point reduction produced when 1 mole of solute is dissolved in 1000 g of the solvent.

(ii) Equimolecular quantities of different substances dissolved in the same quantity of a particular solvent reduce its freezing point to the same extent.

The addition of a nonvolatile solute causes solutions to have higher boiling points than the pure solvent.

- Vapor pressure decreases with addition of non-volatile solute.
- Higher temperature is needed in order for vapor pressure to equal 1 atm.

The change in boiling point is proportional to the number of solute particles present and can be related to the molality of the solution:

$$DT_b = K_b.m$$

where DT_b = boiling point elevation

K_b = molal boiling point elevation constant

m = molality of solution

The answer to your question is provided in the image:

The value of K_b depends only on the identity of the solvent

Problem: Calculate the boiling point of an aquous solution that contains 25.0 mass % ethylene glycol ($C_2H_6O_2$, a nonvolatile liquid).

Problem : The boiling point of an aqueous solution that is 1.0 m in NaCl is 101.02°C whereas the boiling point of an aqueous solution that is 1.0 m in glucose (C6H12O6) is 100.51°C. Explain why.

Solvent	Boiling Point (°C)	K_b (°C/m)	Freezing Point (°C)	K_f (°C/m)
acetic acid	117.90	3.22	16.64	3.63
benzene	80.09	2.64	5.49	5.07
d-(+)-camphor	207.4	4.91	178.8	37.8
carbon disulfide	46.2	2.42	−112.1	3.74
carbon tetrachloride	76.8	5.26	−22.62	31.4
chloroform	61.17	3.80	−63.41	4.60
nitrobenzene	210.8	5.24	5.70	6.87
water	100.00	0.51	0.00	1.86

Our first task is to convert the percent by mass to a molal concentration.

Assume that you have 1000 g of solution.

Then, you have 250 g of ethylene glycol (EG) and 750 g of water.

$$\text{Moles of EG} = 250\ \cancel{\text{g EG}} \times \frac{1\ \text{mol EG}}{62.07\ \cancel{\text{g EG}}}$$

$$= 4.03\ \text{mol EG}$$

$$b = \frac{\text{moles of solute}}{\text{kilograms of solvent}} = \frac{4.03\ \text{mol}}{0.750\ \text{kg}}$$

$$= 5.37\ \text{mol·kg}^{-1}$$

Calculate the boiling point

$$\Delta T_b = K_b b = 0.512\ ^\circ\text{C} \cdot \cancel{\text{kg·mol}^{-1}} \times 5.37\ \cancel{\text{mol·kg}^{-1}}$$

$$= 2.7\ ^\circ\text{C}$$

$$T_b = T_b^\circ + \Delta T_b = 100.0\ ^\circ\text{C} + 2.7\ ^\circ\text{C} = 102.7\ ^\circ\text{C}$$

Calculate the freezing point

$$\Delta T_f = K_f b = 1.86\ ^\circ\text{C} \cdot \cancel{\text{kg·mol}^{-1}} \times 5.37\ \cancel{\text{mol·kg}^{-1}}$$

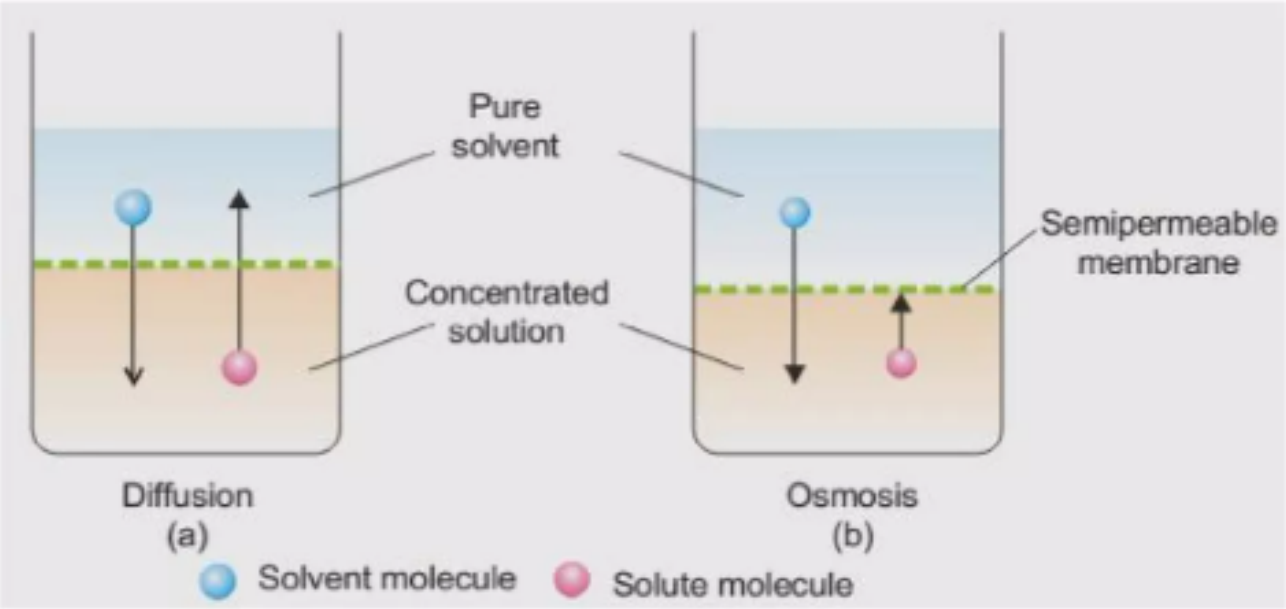
$$= 10.0\ ^\circ\text{C}$$

$$T_f = T_f^\circ - \Delta T_f = 0.0\ ^\circ\text{C} - 10.0\ ^\circ\text{C} = -10.0\ ^\circ\text{C}$$

WHAT IS OSMOSIS?

- Let us consider a pure solvent and solution separated by a membrane which permits the passage to solvent molecules but not to solute molecules. Only the solvent will diffuse through the membrane into solution.

- A membrane which is permeable to solvent and not to solute, is called a **semipermeable membrane**.



(a) Diffusion consists of solute molecules moving upward into the solvent and solvent molecules moving down into concentrated solution;

(b) Osmosis consists of solvent molecules passing through the semipermeable membrane, while large solute particles are held up.

➢ **The flow of the solvent through a semipermeable membrane from pure solvent to solution, or from a dilute solution to concentrated solution, is termed Osmosis (Greek Osmos = to push).**

- It must be clearly understood that the diffusion of solvent molecules through a semipermeable membrane is taking place in both directions.
- That is, solvent molecules are passing from solvent to solution, and also from solution to solvent.
- But since the diffusion from solvent to solution or from dilute to concentrated solution, is more rapid, **the net flow of the solvent is from low to high concentration**.

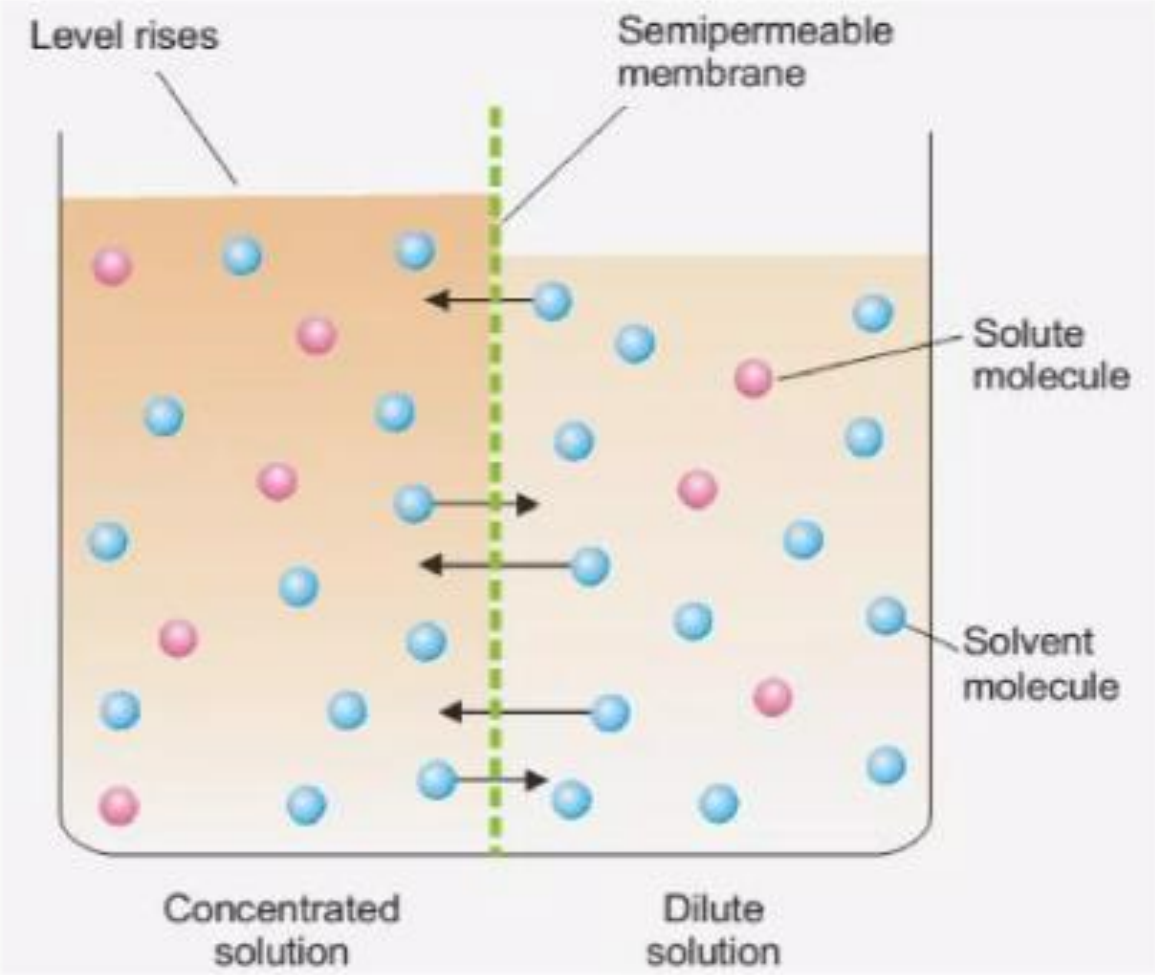
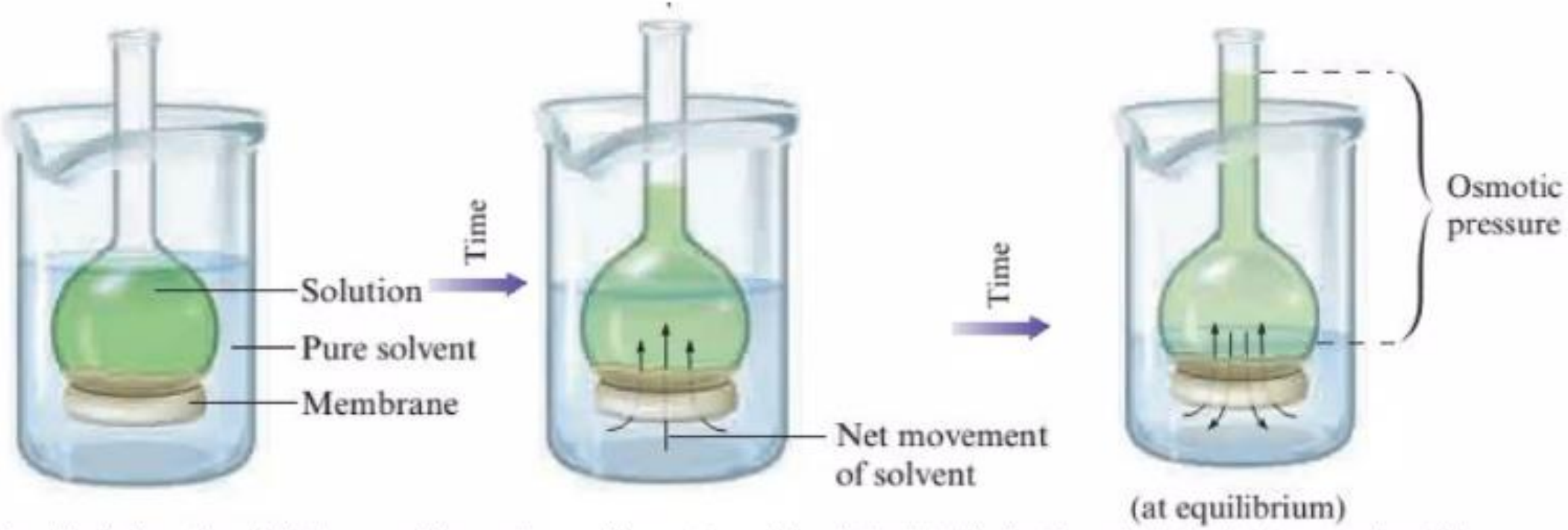


Illustration of the process of osmosis.

- The phenomenon of osmosis is the **spontaneous passage** of a pure solvent into a solution separated from it by a semipermeable membrane, a membrane permeable to the solvent but not to the solute.

- As time passes, the volume of the solution increases and that of the solvent decreases. Eventually the liquid levels stop changing, indicating that the system has reached **equilibrium**.



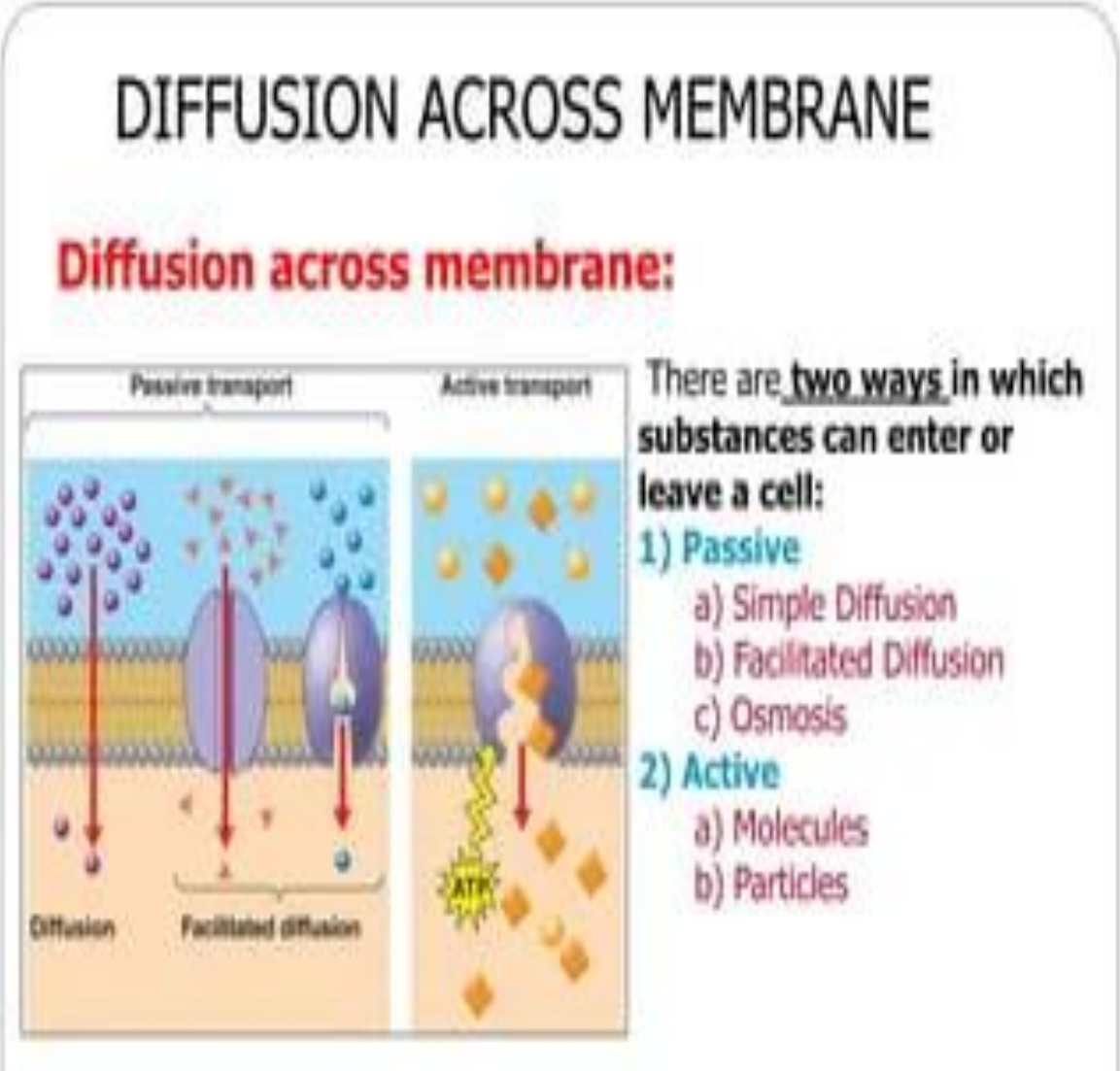
A tube with a bulb on the end that is covered by a semipermeable membrane. The solution is inside the tube and is bathed in the pure solvent. There is a net transfer of solvent molecules into the solution until the hydrostatic pressure equalizes the solvent flow in both directions.

Diffusion

- Diffusion is the movement of a substance from an area of high concentration to an area of low concentration.
- Diffusion occurs in liquids and gases because their particles move randomly from place to place.
- Diffusion is a passive process of transport. A single substance tends to move from an area of high concentration to an area of low concentration until the concentration is equal across the space.
- Diffusion is an important process for living things.

Example of diffusion

- The perfume of a flower that quickly permeates the still air of a room.
- A drop of food coloring diffuses throughout the water in a glass so that, eventually, the entire glass will be colored.
- Such as the movement of oxygen from the air into the blood and carbon dioxide out of the blood into the air in the lungs, or the movement of glucose from the blood to the cells.

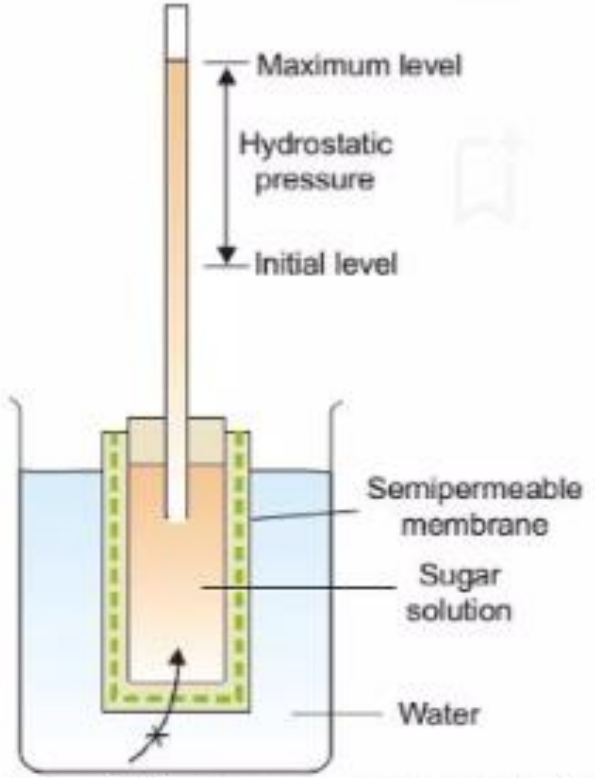


Difference between Osmosis and Diffusion

Osmosis	Diffusion
It is limited only to the liquid medium.	Occurs in liquid, gas and even solids.
Requires a semipermeable membrane.	Does not require a semipermeable membrane.
Depends on the number of solute particles dissolved in the solvent.	Depends on the presence of other particles.
Requires water for the movement of particles.	Does not require water for the movement of particles.
Only the solvent molecules can diffuse.	Both the molecules of solute and solvent can diffuse.
The flow of particles occurs only in one direction.	The flow of particles occurs in all the directions.
The entire process can either be stopped or reversed by applying additional pressure on the solution side.	This process can neither be stopped nor reversed.
Occurs only between similar types of solutions.	Occurs between the similar and dissimilar types of solutions.
It involves the movement of only solvent molecules from one side to the other.	It involves the movement of all the particles from one region to the other.
The concentration of the solvent does not become equal on both sides of the membrane.	The concentration of the diffusion substance equalises to fill the available space.
Depends on solute potential.	Does not depend on solute potential, pressure potential, or water potential.
Only water or another solvent moves from a region of its high concentration to a region of its lower concentration.	Any type of substance moves from area of highest energy or concentration to region of lowest energy or concentration.
Not associated with uptake of minerals and nutrients.	It helps in the uptake of minerals and nutrients.

WHAT IS OSMOTIC PRESSURE?

- A pot with s semipermeable membrane deposited in its walls is fitted with a long glass tube.
- It is filled with concentrated aqueous sugar solution and immersed in distilled water.
- The osmosis of water through the membrane from water to the sugar solution takes place.
- As a result, the solution level in the long tube rises over a period of time. After a few days the level attains a definite maximum value.



Hydrostatic pressure necessary to stop osmosis is equal to osmotic pressure.

- This marks the stage when the hydrostatic pressure set up due to the column of sugar solution counterbalances the flow of pure water (or osmosis) into the solution.

➢ **The hydrostatic pressure built up on the solution which just stops the osmosis of pure solvent into the solution through a semipermeable membrane, is called Osmotic Pressure.**

Van't Hoff's Law of Osmotic Pressure

Quantitative relationship between the concentration of the solution and the osmotic pressure was first derived by Van't Hoff in 1886. These are known as Van't Hoff's laws of osmotic pressure.

First Law: The osmotic pressure of a solution at a given temperature is directly proportional to its concentration.

If π is the osmotic pressure and C its concentration in mole/L, we can write $\pi \propto C$, if temperature is constant.

$\pi \alpha C$ at constant T ----------(i)

If V is volume containing one mole of solute, then C=1/V (since concentration, C=mole/Volume)

Thus, $\pi \alpha 1/V$ at constant T

Or, πV = constant

Second Law: The osmotic pressure of a solution of a given concentration is directly proportional to the absolute temperature.

If T is the absolute temperature, we can write

$\pi \propto T$, if concentration is constant ----------(ii)

Third Law: Equimolecular quantities of different solutes dissolved in such volumes of solvent as to give the same volume of the solution have the same osmotic pressure at the same temperature.

Combining equation (i) and (ii) ---

$\pi \alpha T/V$

$\pi V = RT$ (for one mole of solute/V liter of solution)

$\pi V = nRT$ (for n mole of solute/V liter of solution)

$\pi V = w/m.\ RT$ [w is weight in gm and m is MW]

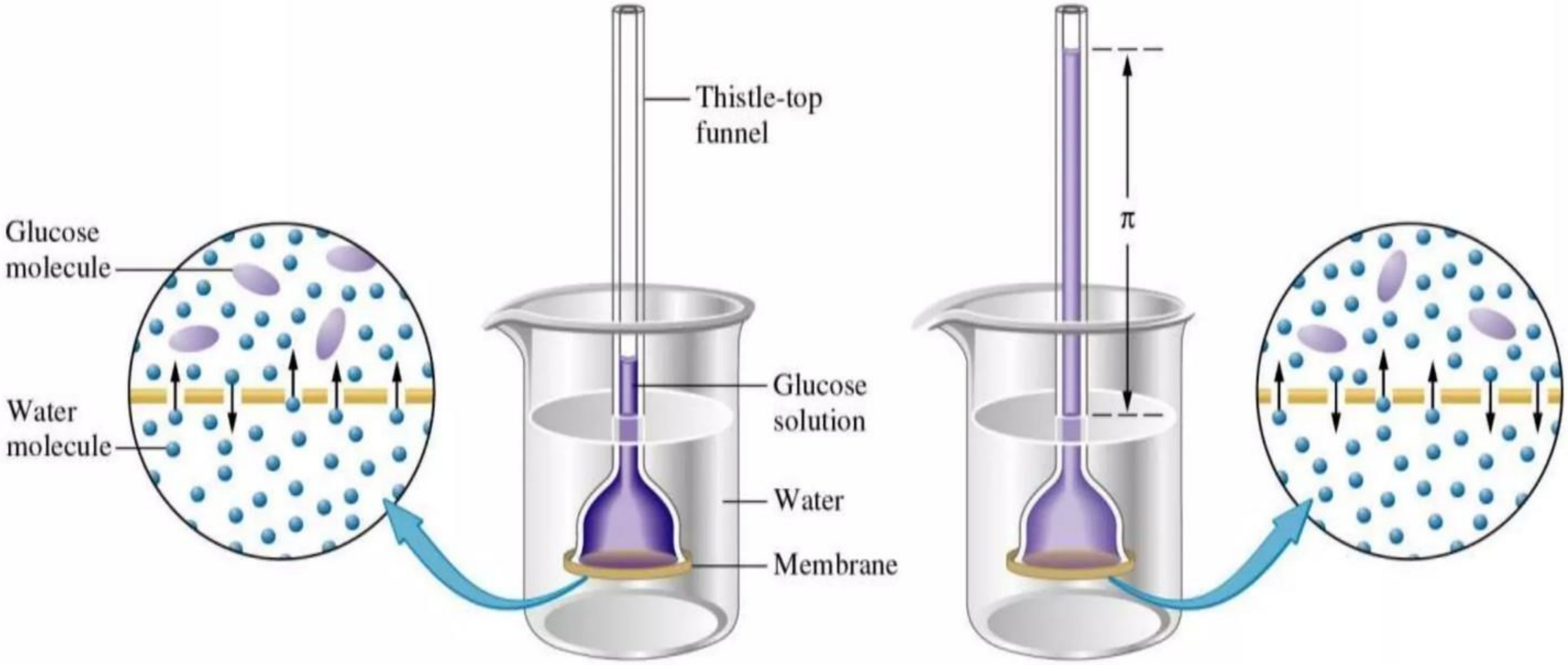
Determination of MW from Osmotic Pressure

$\pi V = w/m.\ RT$ [w is weight in gm and m is MW]

$m = wRT/\pi V$

Level rises

Semipermeable membrane



solution

solution